

CLOUD COMPUTING COURSE

Lecture8-Compute Virtualization

ABSTRACT

Lecture8 Introduces compute virtualization technology.

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Compute Virtualization

1. Compute Virtualization Definition:

It is a technique, which allows sharing single physical instance of a resource among multiple organizations or tenants (customers). It does so by **assigning a logical name** to a physical resource and providing **a pointer to that physical resource** on demand.

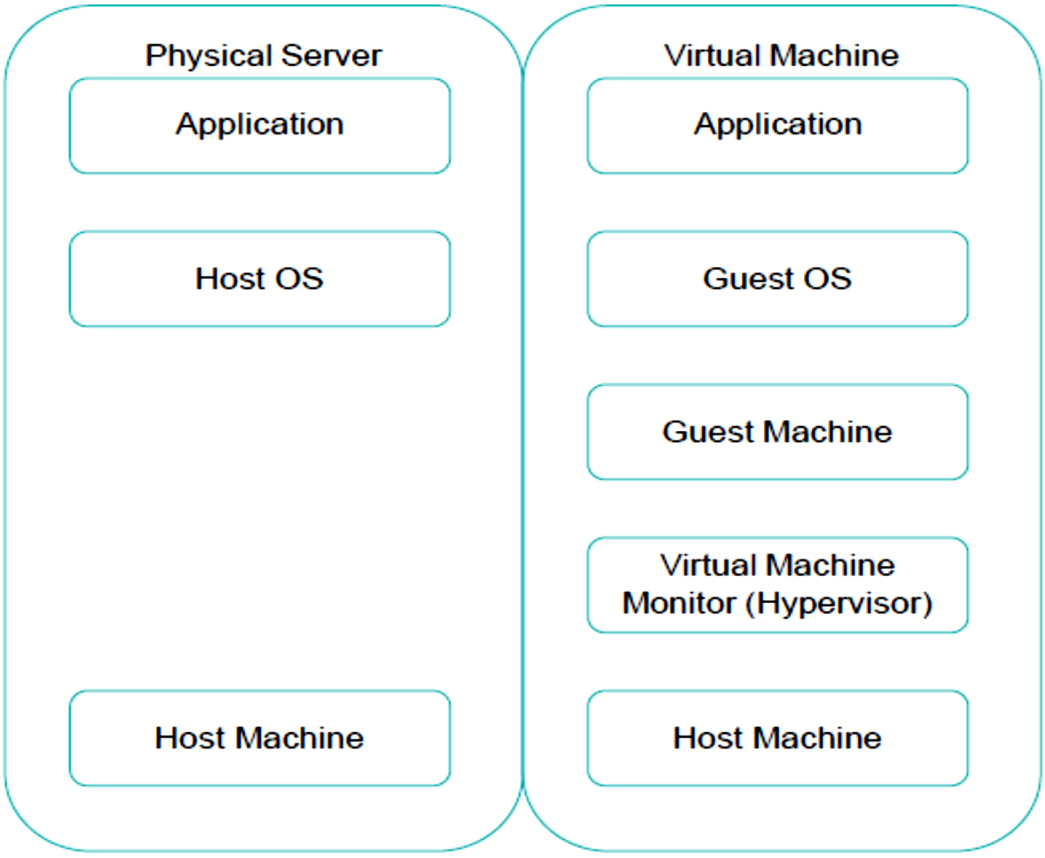
This done by using a software layer called a **hypervisor**. The resources includes the **CPU's, memory, input/output (I/O) and network traffic.**

Hypervisors take the physical resources and separate them so they can be utilized by the virtual environment.

2. Virtualization Concept

Creating a virtual machine over existing operating system and hardware is referred as Hardware Virtualization. Virtual Machines provide an environment that is logically separated from the underlying hardware.

The machine on which the virtual machine is created is known as **host machine** and virtual machine is referred as a **guest machine**. This virtual machine is managed by a software, which is known as **hypervisor**.



Guest OS:

Operating system running in a virtual machine (VM)

Guest Machine:

Virtual machine created through virtualization

Hypervisor:

Virtualization software layer, or

Virtual Machine Monitor (VMM)

Host OS:

Operating system running in a physical machine

Host Machine:

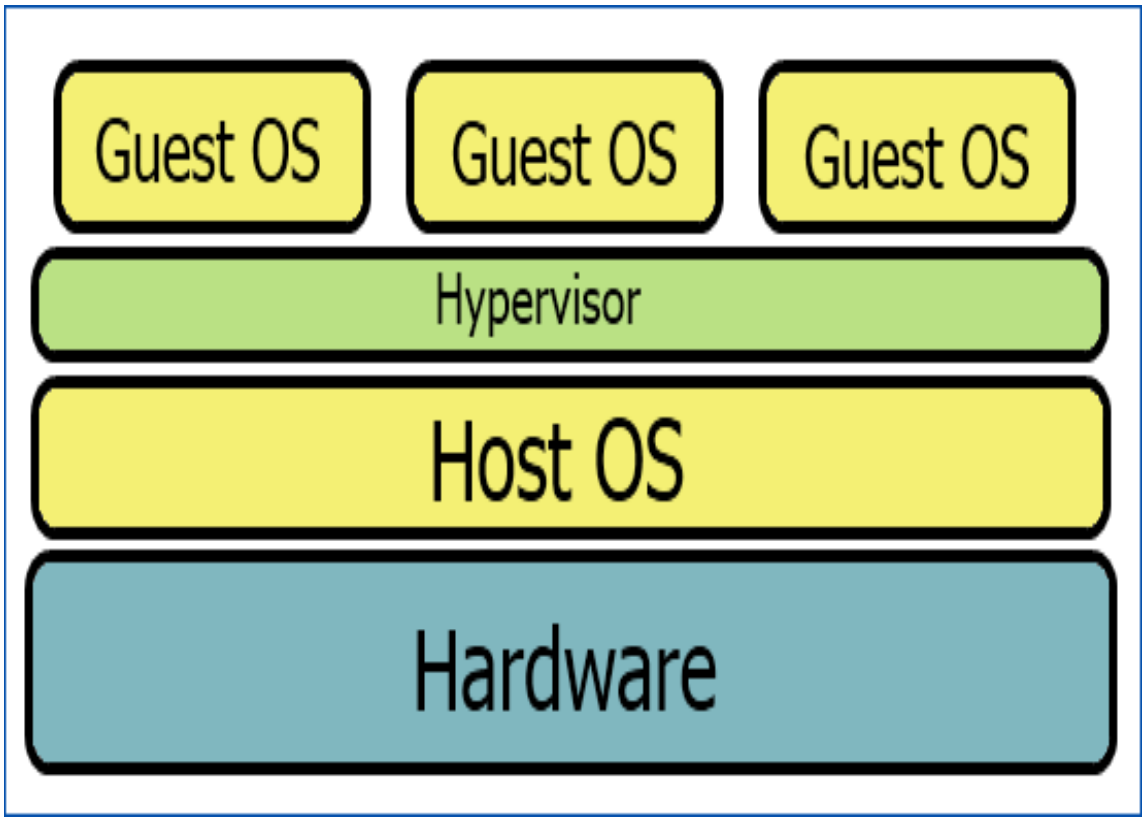
Physical machine

VM Definition1: A virtual machine is a computer file, typically called an image that behaves like an actual computer. In other words, creating a computer within a computer. It runs in a window, much like any other program, giving the end user the same experience on a virtual machine, as they would have on the host operating system itself.

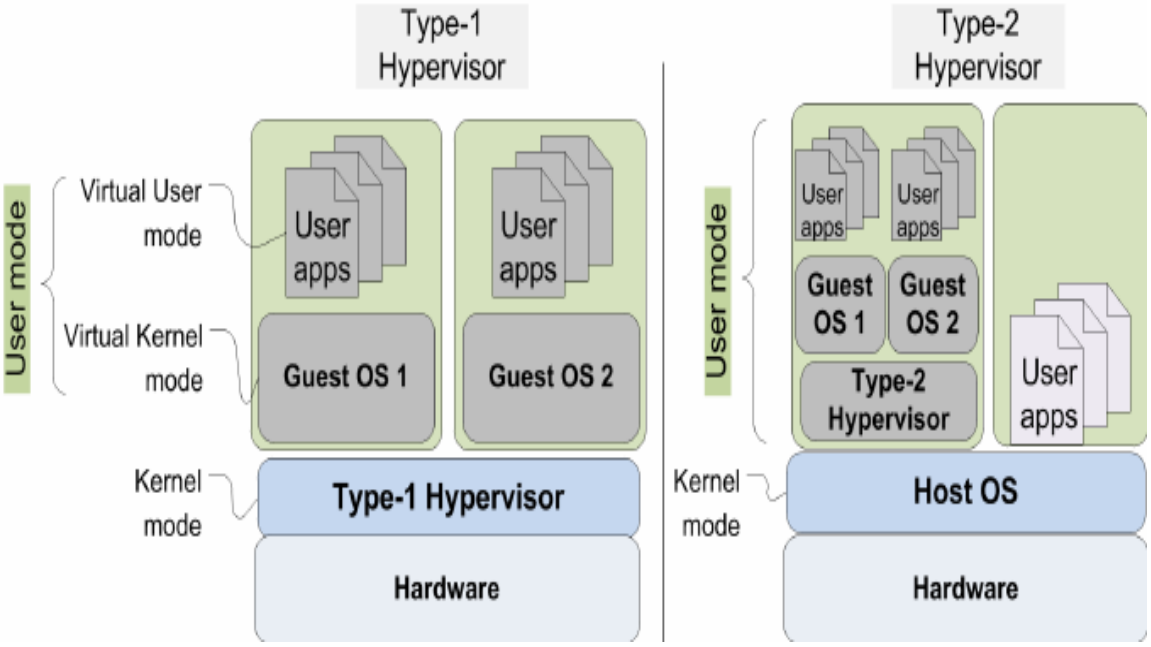
VM Definition2: A tightly isolated software container with an operating system and application inside. Each self-contained VM is completely independent.

While virtualization is converting physical servers into logical folders or files called **VM**'s, these folders or files (**VM**) divided into two parts:

1. VM configuration files
2. User data files



3. Hypervisor Types:

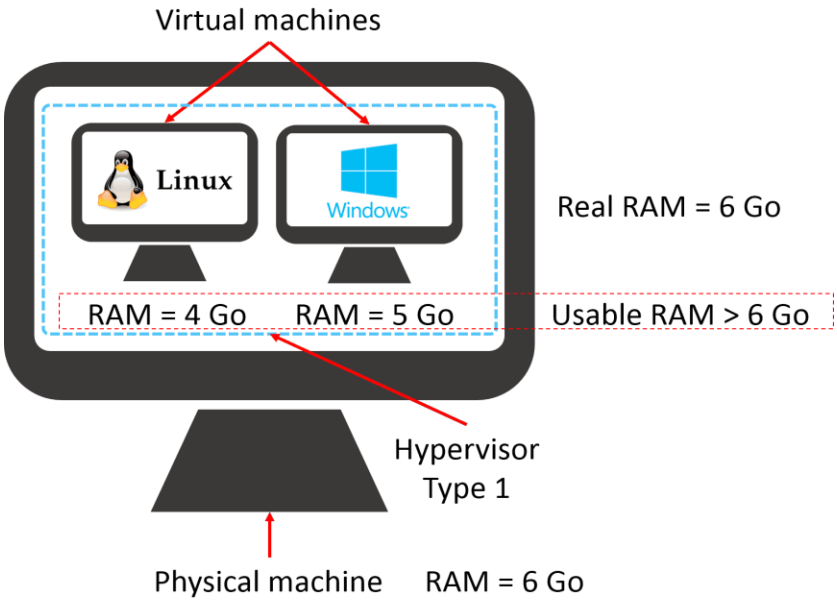


Hypervisors are able to sit on top of an OS or they can be directly installed onto the hardware. This direct us to the different types of hypervisor, which comes in two types:

1. **Type 1** (also called **bare-metal**): The most common hypervisor in **cloud computing**, it is designed to sit directly on underlying hardware and provide the ability to virtualize the hardware resources for use by the virtual machines.

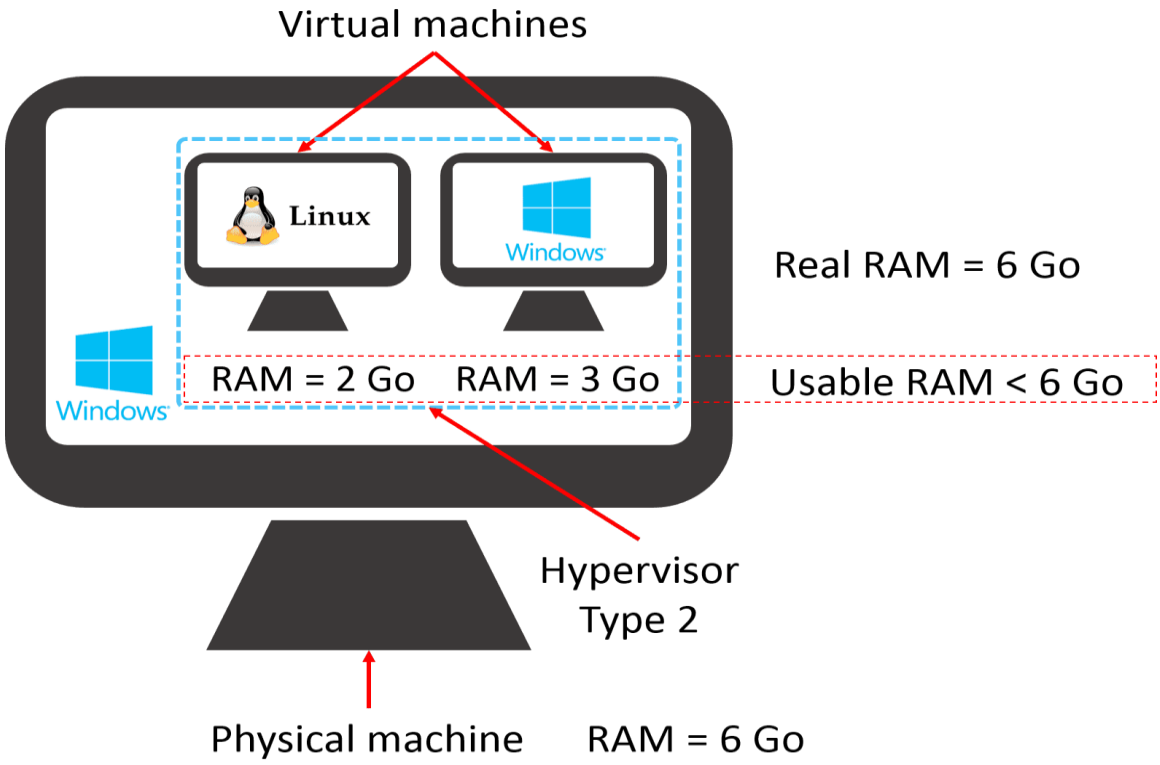
Type1 Hypervisor Examples:

1. VMware ESXi
2. Citrix XenServer
3. Microsoft Hyper-V



2. **Type 2** hypervisor (also **called hosted**): requires a host operating system and it runs on this host machine as another applications and used to create VM's behind the host own applications.

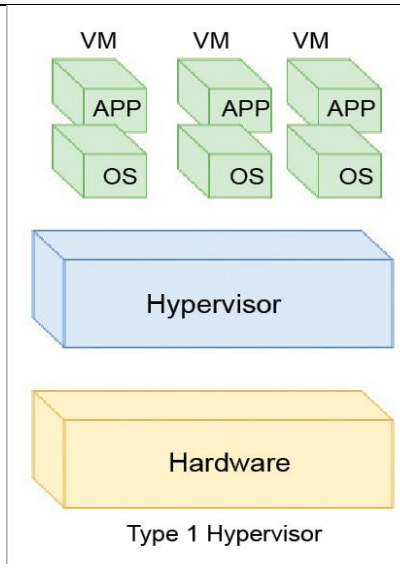
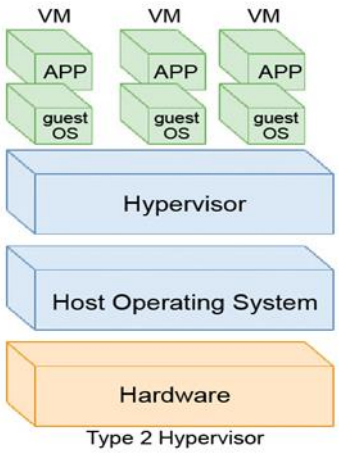
It is more often used for **testing and labs and for home usage.**



Type2 Hypervisor Examples:

1. VirtualBox
2. Qemu
3. VMware Workstation player.

Comparison between Type1 and Type2

Feature	Type1	Type2
Installation Position	Run directly on the system Hardware.	Run on a host OS.
Support	Hardware Virtualization	OS Virtualization
Security	More secure	Less secure
Performance	Very high	Less than type1
Advantage	VMs can run different types of guest OSs and applications independent of the host OS.	Easy to implement
Disadvantage	Hard Development	Only the applications supported by the host OS can be installed and used
Example	VMware ESXi , XEN Server	KVM, Virtual Box
	 Type 1 Hypervisor	 Type 2 Hypervisor

4. Virtualization Features

There are four main **VM features** supported by both type1 and type2 hypervisors:

1. **Partitioning**
   - Run multiple operating systems on one physical machine.
   - Divide system resources between virtual machines.

2. **Isolation**
   - Provide fault and security isolation at the hardware level.
   - Preserve performance with advanced resource controls.

3. **Encapsulation**
   - Each VM is saved as a group of hardware-independent files.
   - Move and copy virtual machines as easily as moving and copying files, which support VM migration.

4. **Hardware Independence**
   - Provision or migrate any virtual machine to any physical server.
   - The migration of VMs can be successful as long as **the same VMM** running on the **target host** as that on the **source host**.

5. Virtualization vs. Cloud Computing

Virtualization is software that makes computing environments independent of physical infrastructure, while **cloud computing** is a service model that delivers shared computing resources (software and/or data) on demand via the Internet. As complementary solutions, organizations can begin by virtualizing their servers and then moving to cloud computing for even greater agility and self-service.

Virtualization and cloud computing **are not interchangeable.**